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Development of hydrophobic surface substrates enabling reproducible drop-and-dry spectroscopic measurements

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ABSTRACT

We investigated several spectroscopic substrates with hydrophobic surfaces that were able to form reproducible droplets of aqueous samples for reliable high throughput drop-and-dry measurements. An amine-coated substrate, a polytetrafluoroethylene (PTFE) disk, and a perfluorooctyltrichlorosilane (FTS) coated substrate were prepared and initially evaluated for use in the determination of fat concentrations in milks using near-infrared (NIR) spectroscopy. Since the dried milk spots were not compositionally uniform due to the localization of components during sample drying, NIR spectra were collected by fully covering each spot to ensure a correct compositional representation of the sample. The amine-coated substrate yielded more reproducible dried milk patterns because its hydrophobicity was optimal for loading an appropriate amount of milk with decreased component localization after drying. The relative standard deviation (RSD) of the absorbance at 4330 cm^{-1} was 1.0%, thereby resulting in the more accurate determination of fat concentration. In addition, infrared (IR) spectroscopic discrimination between wild and transgenic tobaccos using their extracts was attempted. The extracted metabolites had a low concentration, so an FTS-coated CaF_2 substrate that maximized sample loading was used to improve measurement sensitivity and produce reproducible droplets. The RSD of the absorbance at 1070 cm^{-1} was only 0.8%. Our strategy produced droplets that had consistent sizes and provided reproducible IR spectral features, which enabled the differentiation between wild and transgenic tobacco groups in the principal component (PC) score domain.

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1. Introduction

Near-infrared (NIR) spectroscopy is a mature analytical method used for fast non-destructive analysis of diverse samples [1–7]. High throughput NIR analysis [8–10], which measures a large number of samples rapidly, has become popular for routine analysis. When a large number of aqueous samples are subject to analysis, one effective method is the so-called drop-and-dry method, which is the rapid, automated spectral acquisition of multiple dried samples formed on a substrate after solvent evaporation [11–13]. Many samples can be quickly measured using this method and, more importantly, the problem of strong water absorption that limits the available spectral window for NIR analysis and eventually degrades the signal-to-noise ratio of spectra can be eliminated.

Reliable drop-and-dry measurements require the generation of reproducible sample shapes after drying [14–18]. If the shapes of the dried samples are inconsistent, the reproducibility of the observed spectral features and the subsequent accuracy of analysis will be degraded. Typically, a 96-well plate holding a sample within a boundary is used with encompassing walls. However, a loaded sample is attracted to the edge of the well due to surface adhesion, so dissolved components tend to be localized along the wall during solvent evaporation. Component aggregation along the wall makes the acquisition of representative spectra difficult.

In this publication, we have investigated several drop-and-dry substrates that can be used to generate consistent droplet shapes without using any physical partitioning walls for NIR and infrared (IR) spectroscopic analysis. When aqueous samples are dropped on a common substrate such as a glass slide possessing a relatively hydrophilic surface, the droplet spreads out due to the sample-surface interaction, so it is difficult to obtain reproducible droplet shapes. One effective alternative is the use of a surface with a polarity opposite to that of the aqueous sample, which serves to confine the sample within a given area with a regular shape. To evaluate this strategy, three different substrates possessing hydrophobic surfaces (a polytetrafluoroethylene (PTFE) disk,

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perfluorooctyltrichlorosilane (FTS) coated glass and amine-coated glass) were prepared and used for drop-and-dry NIR measurements of milk to determine fat concentration. Milk samples were dropped on the substrates, and the corresponding contact angles and reproducibility of droplet formation were compared.

Another important issue for reliable drop-and-dry measurements is the acquisition of spectra representative of the whole dried spot. This is important because the distribution of the dried components in a sample is not always uniform due to the dissimilar interactions between the components and the surface during solvent evaporation. A spectrum covering only a part of the dried sample fails to represent the whole composition of the sample. Therefore, NIR spectra were collected by fully covering each dried milk sample with a proper optical arrangement to ensure representative sampling. The measured spectra were used to determine fat concentrations using a partial least squares (PLS) method.

To extend the applicability of the scheme further, IR spectroscopic discrimination of wild and transgenic tobaccos (genetically modified to have a great chloroplast) was attempted as a simple example of metabolic fingerprinting. Metabolic fingerprinting, which is a sub-field of metabolomics, attempts to classify large numbers of samples based on chemical analysis; it can be used to elucidate different metabolic pathways, and so it requires efficient high throughput analytical methods [19,20]. Unlike the measurements on milk, the concentrations of the extracted metabolites from tobacco samples were low, so an FTS-coated CaF_2 substrate possessing a strong hydrophobic surface was used to maximize sample loading for sensitivity and to provide reproducible sample droplets before drying. The distribution of the principal component (PC) scores obtained from IR spectra of the samples dried on the substrate was examined to identify wild and transgenic tobaccos.

2. Experimental section

2.1. Milk and tobacco samples

A total of 90 commercially available milks from 10 different manufactures were purchased at local markets. The milk fat concentration was 0.0%, 2.0% or 4.0%. To prepare samples with different fat concentrations, two samples were mixed to generate concentrations in the 0.0–4.0% range at a concentration interval of approximately 0.09%. A total of 45 milk samples were prepared.

Wild and transgenic tobacco (*Nicotiana tabacum* cv. Xanthi) plants were grown in constant environment conditions at 25 °C on a 16-h photoperiod in a growth chamber (Conviron, Plant Growth Chamber CG108). The transgenic great chloroplast tobaccos were generated by introduction of the *NTFtsZ* gene [21,22]. The relative humidity was not controlled, but was maintained within the range of 60-80%. The transgenic tobaccos showed similar growth rates in moderate light conditions compared to the wild tobaccos; however, their growth was greatly retarded under high or low light intensity. Approximately eight-week-old plant leaves were collected and washed with distilled water. After removal of the remaining moisture with filter paper (3 M), the leaves were immediately frozen with liquid nitrogen and then freeze-dried. After being ground to a fine powder, leaf samples were stored at −70 °C. Finally, 13 wild and 13 transgenic tobacco leaves were prepared. Thirty milligrams of a powder sample was added to 600 mL of an extraction solvent (water (80%):methanol (20%)) in an Eppendorf tube. Then, the tube was sonicated for 90 min at 40 °C and was centrifuged for 5 min at 13,000 rpm. The supernatant solution was used for measurement.

2.2. Preparation of substrates

Amine-coated glass, originally designed for attachment of negatively charged phosphate groups in a DNA backbone for microarray analysis, was purchased from LumiNano (Seongdong-Gu, Seoul, Korea). A polytetrafluoroethylene (PTFE) disk (thickness: 0.5 mm) was purchased at a local store. For the preparation of the FTS-coated glass substrate, a clean glass slide was activated with UV–O_3 to generate hydroxyl groups on the surface for the coating, and the slide was kept in a vacuum chamber under 20 mtorr with FTS overnight at room temperature. An FTS-coated CaF_2 substrate was prepared by the same procedure for the IR spectroscopic measurement of the tobacco extracts. Contact angles of the samples on these substrates were measured using an automatic contact angle measuring system (Easydrop DSA20S, KRÜSS, Germany).

2.3. Acquisition of NIR and IR spectra

To acquire representative sample spectra by fully covering a substrate with a dried sample, the sampling part of a spectrometer (ABB Bomem Inc., Canada) was modified for collimating radiation using a concave lens to form a large illumination area up to 100 mm^2 (10 mm × 10 mm). An Al plate with a hole matching the size of a dried spot was positioned for measurement (128 scans). Dried milk and tobacco extract spots were 10 and 6 mm in diameter, respectively. The resolutions of NIR and IR spectra were 8 and 4 cm^{-1}, respectively.

NIR images of the dried milk were acquired using a Hyperion 3000 FT-UV–vis-IR microscope attached to a Bruker Vertex 80 spectrometer equipped with a MCT detector. The mapping interval was 580 μm in both the *x* and *y* directions (aperture size: 100 μm), and a total of 400 spectra were collected in each mapping. IR line mapping of a dried spot of tobacco extract was performed using an IR microscope (aperture size: 50 μm) equipped with an MCT detector (SensIR Technologies, Danbury, CT, USA). All the spectral pretreatments and multivariate analyses, including baseline corrections, normalizations, PLS and principal component analysis (PCA), were accomplished using Matlab version 7.0 (The MathWorks Inc., MA, USA). NIR images of the dried milks were also constructed using the same software.

3. Results and discussion

3.1. Examination of droplet shapes formed on substrates

The surface hydrophobicity influences the droplet shape and volume formed on a substrate. For an aqueous sample, a more hydrophobic surface results in a droplet that is more condensed at the surface, which could help to provide more consistency in droplet shape. To examine the formation of droplets on a glass slide as well as the tested substrates, contact angles of milk droplets (fat concentration: 4.0%) were measured. The volume of each droplet was 150 μL. The FTS surface produced the highest contact angle of 102.1°, and the contact angle decreased to 84.4° for the PTFE disk. The strong hydrophobicity of these substrates was responsible for the high contact angles. As expected, the amine surface further decreased the contact angle to 50.2°, and the contact angle was smallest on the bare glass surface (17.2°).

When the same sample volume was loaded, the contact area of the droplet decreased with increasing surface hydrophobicity. Once a target spot size was determined, a larger sample volume can be loaded on a more hydrophobic surface, which improves the measurement sensitivity. The size of the droplet contact area for analysis of milks was set to 10 mm in diameter, and the loading volumes of milk required to achieve this diameter were

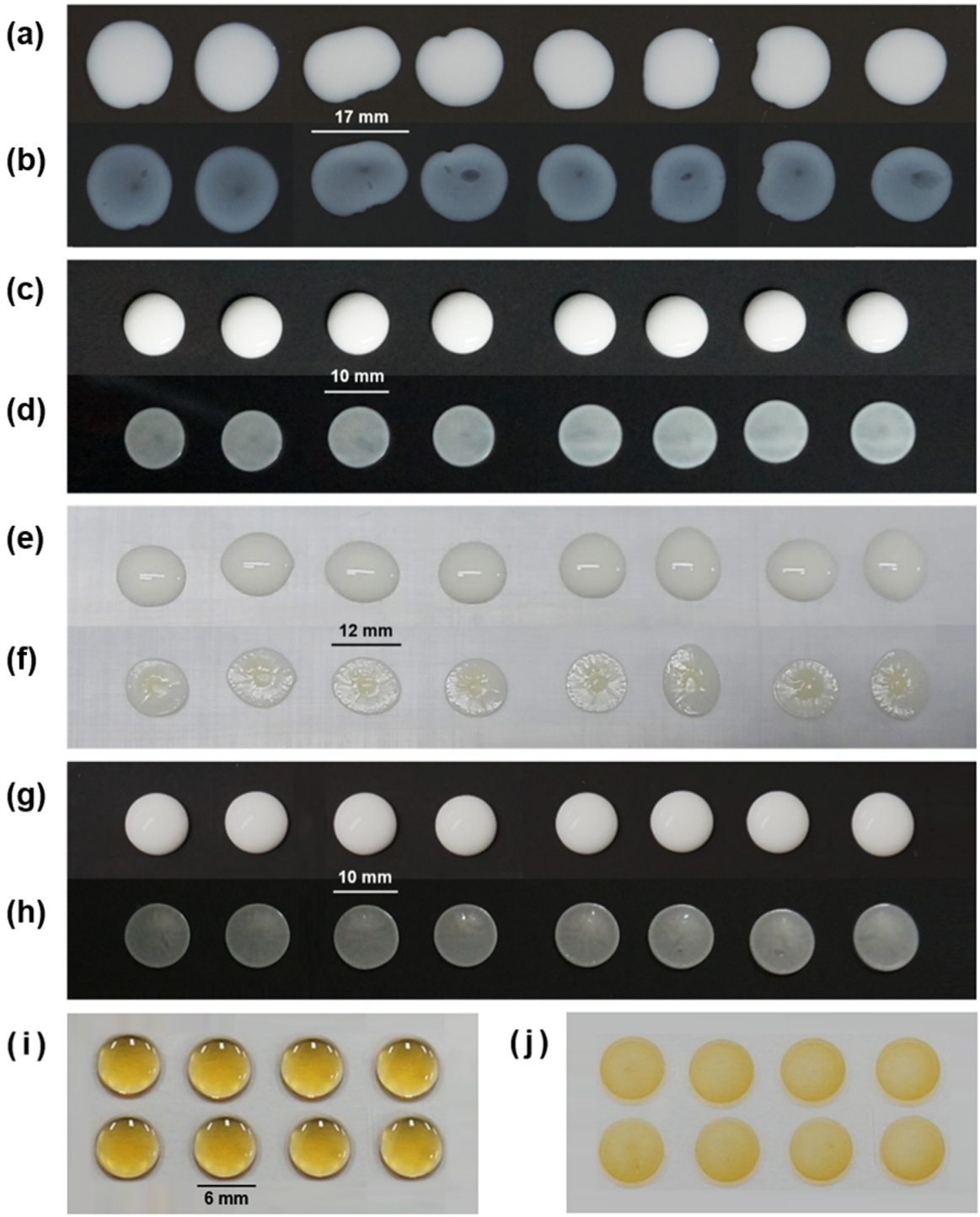

Fig. 1. Top-view pictures of 8 replicate droplets of 4.0% fat milk on (a) the glass slide, (c) amine-coated substrate, (e) PTFE disk, and (g) FTS-coated substrate, respectively. The pictures of the corresponding dried samples are shown just below each picture. Top-view pictures of 8 replicate droplets of a tobacco extract (i) and the corresponding dried spots (j) are also shown.

determined for each substrate. The determined loading volumes were 100, 150, 200 μL for the amine-coated substrate, PTFE disk and FTS-coated substrate, respectively, as shown in Fig. 1. Even 50 μL of sample created a droplet larger than 10 mm on the slide glass.

To examine the reproducibility of droplet formation, 8 replicate droplets of 4% fat milk were placed on each substrate, and their dried shapes were compared. Fig. 1(a), (c), (e) and (g) shows top-view pictures of 8 replicate droplets on the glass slide, amine-coated glass, PTFE disk and FTS-coated glass, respectively. The pictures of the corresponding dried samples are shown just below each picture of the sample droplets. The droplet shapes on the slide glass were inconsistent, and so the shapes after drying were different from one another. In contrast, the droplets and their dried spots were very consistent for both the amine- and FTS-coated substrates. The increased surface hydrophobicity confined the droplets within a given area and resulted in reproducible droplet formation. Unexpectedly, the droplet shapes formed on the PTFE disk varied even though the surface was more hydrophobic than the amine surface. An examination of the PTFE surface

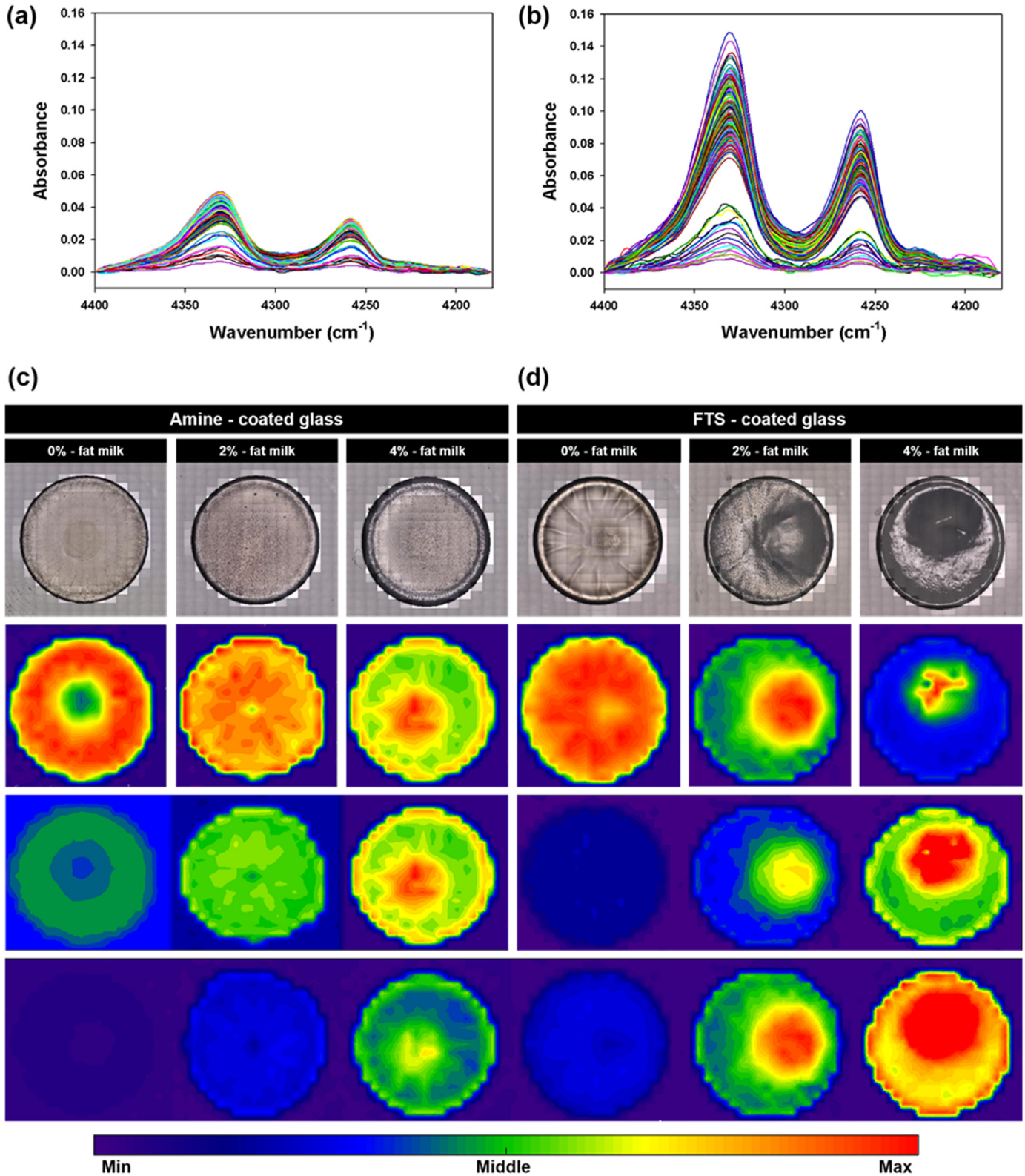

Fig. 2. NIR spectra (4400–4180 cm^{-1} range) mapped within the dried samples of 4.0% fat milk on (a) the amine- and (b) FTS-coated substrates. A total of 250 spectra are displayed in each case. The visual (1st row) as well as score-based images (2nd to 4th row) of dried 0.0%, 2.0% and 4.0% fat milks on (c) the amine- and (d) FTS-coated substrates are also shown.

using a microscope showed that it had fine grains and was not as smooth as a glass-based surface. The grained surface eventually hampered the formation of shape-consistent droplets. Therefore, the slide glass and PTFE disk were excluded in further measurements.

3.2. Microscopic examination of a dried milk spot

As seen in Fig. 1(h), the center of each dried sample on the FTS-coated substrate was more condensed, and there was compositional inhomogeneity within the sample. NIR mapping (point

spacing: 580 μm, 20 × 20 points) was performed over a dried sample and the variation of the spectral features among the mapped spectra was examined. Three dried milk samples of 0.0%, 2.0% and 4.0% fat were mapped on both the amine- and FTS-coated substrates. Fig. 2(a) and (b) shows the NIR mapping spectra (4400–4180 cm^{-1} range) acquired from the dried 4% fat milk on the amine- and FTS-coated substrates, respectively. The baselines of the spectra were corrected at 4400 and 4180 cm^{-1}. Out of a total of 400 spectra in each map, 250 that were acquired within the sample spot are shown. The peaks at 4330 and 4260 cm^{-1} correspond to the vibrational combinations associated with the hydrocarbon chains in fatty acids [23]. The magnitudes of absorption were larger for the sample on the FTS-coated substrate due to the greater sample loading. In both cases, the absorbances varied significantly, and the fat concentrations in the dried samples were clearly not uniform.

For further examination, mapping images of the dried samples were constructed using the first scores obtained by performing PCA. Fig. 2(c) and (d) shows the visual (1st row) and score-based images (2nd to 4th row) of the dried 0.0%, 2.0% and 4.0% fat milks on the amine- and FTS-coated substrates, respectively. The gray tone is not uniform across the spot in the visual images of each sample. Localized inner circles with a dark gray color are apparent in the images of the 2.0% and 4.0% fat milks on the FTS-coated substrate.

Each score image in the second row was constructed by separately incorporating only the mapping spectra of each sample. This was to maximize the contrast in the score variation within a dried sample. In the image of the 0.0% fat milk on the amine-coated substrate, there is an internal circular area with green-blue/blue colors, and the corresponding band intensity in this region is lower than that of the outer ring area. A droplet becomes smaller due to solvent evaporation, and then dissolved components start to dry from the edge to the center of the droplet [24–27]. In the case of the 0.0% fat milk, the quantity of dissolved components such as protein and lactose is insufficient to cover the droplet area, so a central circular area is formed with few or no dried components. In the image of the 2.0% fat milk, the central circular area disappears due to the full surface coverage by the increased fat concentration, and the color distribution is relatively consistent. When the fat concentration increases to 4.0%, a red/orange circle corresponding to high fat absorption begins to occur inside the spot. Since the fat concentration is high, the evaporation makes the dissolved fat more concentrated in the middle.

The score images of the dried samples on the FTS-coated substrate are different. In the image of the 0.0% fat milk, the circular ring observed on the amine surface completely disappears due to the increased surface coverage from the doubled sample loading, and the color distribution is fairly uniform. The center-condensed image is observable even for the 2.0% fat milk and becomes more dominant for the 4.0% fat milk. It is important to note that fats in milk exist in the form of fat globules with sizes ranging from 0.1 to 10 μm in diameter, and they are much larger than other major components. Therefore, as shown in previous reports [28–30], larger particles such as fat globules tend to aggregate at the center during solvent evaporation.

The score images on the third row were constructed by incorporating the mapped spectra corresponding to 3 dried samples of 0.0%, 2.0% and 4.0% fat milks on each substrate. The images were constructed to easily visualize the concentration change of the samples on each substrate. The score images on the fourth row were constructed using all the sample spectra acquired on both substrates. In the cases of 0.0% fat milks on both substrates, the dried shapes are barely observable due to the low NIR absorption. Also, the samples on the FTS-coated substrate have a higher NIR absorption compared to those on the amine-coated substrate as indicated by the colors in the respective images.

3.3. Collection of representative spectra for dried milk samples and the determination of fat concentration

As shown in Fig. 2, the milk components tend to localize during drying, and the dried patterns vary depending on the fat concentration. Therefore, it is essential to acquire composition-representative spectra by illuminating NIR radiation over the whole dried sample for accurate quantitative analysis. For this purpose, collimated NIR radiation with a large illumination area (100 mm^2, 10 mm × 10 mm) was applied to fully cover each dried spot for spectral collection. Six replicated droplets of 2.0% and 4.0% milk were formed on both amine- and FTS-coated substrates, and the corresponding NIR spectra were acquired for the evaluation. Interestingly, the spectral features were more reproducible on the samples on the amine-coated substrate were measured. For the 4.0% fat milk, the relative standard deviations (RSDs) of the absorbance at 4330 cm^{-1} (one of the main absorption bands as shown in Fig. 2(a)) were 1.0% and 3.9% for the amine- and FTS-coated substrates, respectively.

In the case of dried 4.0% fat milk on the FTS-coated substrates, a shape like a volcano with a considerable height appeared in the sample (refer to Fig. 1) due to the localization of the components around the center during sample drying. The volcano-shape could alter the propagation of NIR radiation considerably via both scattering and refraction, so causing fluctuations in the number of photons reaching the detector. Ultimately this would result in poor reproducibility in the measurement. In addition, the locations of the center mounts were not exactly centered, and their shapes were slightly different from each other. In contrast, the degree of component condensation around the middle for the same sample on the amine-coated substrate was smaller as observable in the score image (Fig. 2). A strong hydrophobic surface producing a high contact angle was advantageous for increasing sample loading; however, it might not always be desirable when aqueous samples containing components with high concentrations such as milk are analyzed. The increased sample loading made the dissolved components highly condensed in a localized area after drying, which produced spectra with degraded reproducibility as described earlier. Therefore, the less hydrophobic amine surface was more useful for decreasing the degree of component localization for milk samples and obtaining more reproducible spectral features.

To build a PLS model to determine the fat concentrations, NIR spectra (4400–4180 cm^{-1} range) of 45 milk samples (0.0–4.0% fat) dried on the amine-coated substrate were used. The optimal number of factors was determined by 5-fold cross-validation, and the obtained standard error of calibration (SEC) was 0.12% with using 6 factors. To evaluate the prediction performance of the model, two separately constructed datasets, including 45 newly prepared samples with the same concentration range in each dataset, were used. The NIR spectra of the samples in each dataset were acquired at different periods. The standard error of prediction (SEP) acquired from prediction of the fat concentrations of all 90 validation samples was 0.12%. Fig. 3 shows the concentration correlation plot for the determination of fat concentrations in the milk samples. Open and filled circles correspond to the calibration and prediction data, respectively. As shown, the fat concentrations were accurately predicted without any indication of prediction bias and/or slope. The achieved accuracy was comparable to that reported in previous researches in which NIR measurements were performed on liquid milk [28–31]. For comparison, the same measurement was repeated using the FTS-coated substrate, and the resulting SEC and SEP were degraded to 0.25% and 0.30%, respectively, as expected.

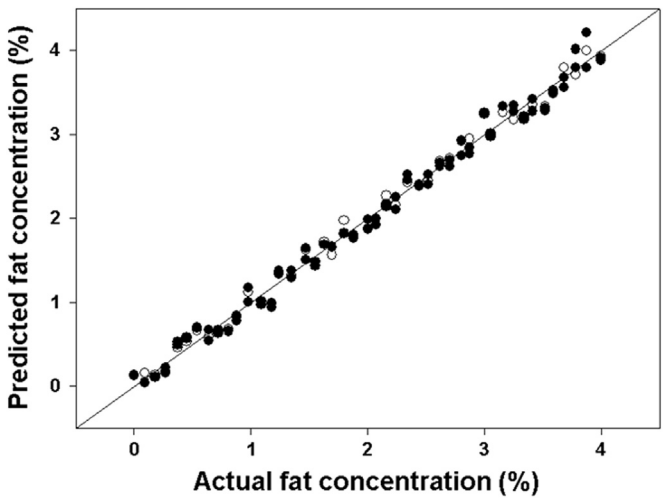

Fig. 3. Concentration correlation plot for the determination of fat concentrations in milks. Open and filled circles correspond to the calibration and prediction data, respectively.

3.4. IR spectroscopic discrimination of wild and transgenic tobaccos

For drop-and-dry measurements using IR spectroscopy, an IR-transparent substrate that allowed for reproducible droplet formation was initially required. Therefore, an FTS-coated CaF_2 substrate prepared using a chemical vapor deposition (CVD) method was used as described above. The concentrations of the extracted metabolites in the samples were very low, so a strong hydrophobic surface such as FTS was required to increase the sample loading and sensitivity of the spectroscopic analysis. To observe the shapes of the droplets, an extract solution from a wild tobacco sample was separately dropped 8 times (200 μL per drop) on the substrate and was then allowed to dry under ambient conditions. Fig. 1(i) and (j) shows the picture of the replicate droplets and the corresponding dried spots, respectively. The shapes of droplets (diameter: 6 mm) and their dried spots are very reproducible.

Since the composition of the extracted components was complex and their chemical structures were highly diverse, the formation of successive ring patterns, the so-called coffee ring effect [32–33], was expected. This effect is the result of dissimilar interactions of components with surfaces over the course of solvent evaporation. IR spectra were collected along a diameter of one dried spot with an interval of 80 μm (aperture size: 100 μm). A total of 73 spectra were collected during the line mapping, and then PCA was performed using the mapped spectra in the 1750–950 cm^{-1} range. The baselines of the spectra were corrected at both 1750 and 950 cm^{-1} before performing PCA. Fig. 4(a) presents the variation in the first (solid line) and second scores (dashed line) along the diameter of the dried spot. Spot numbers 1 and 73 correspond to both edges, and the center is at spot number 37. As shown, the scores continuously vary along the mapping line, and the variation of scores is essentially symmetric with respect to the center. This result clearly shows the formation of successive rings of dried components. To fully represent all the extracted metabolites from a sample, IR spectra were also collected by obtaining a spectrum of an entire dried spot. Fig. 4(b) shows IR spectra collected from the 8 dried spots shown in Fig. 1(j). The spectral features were reproducible, and the RSD of absorbance at 1070 cm^{-1} was 0.8%, which is similar to that obtained in the measurement of milk samples on the amine-coated substrate.

Fig. 5(a) shows the average IR spectra of the wild and great chloroplast tobacco samples. The average spectrum of the wild tobacco samples was arbitrarily offset for comparison. Both spectral features were somewhat different from each other throughout

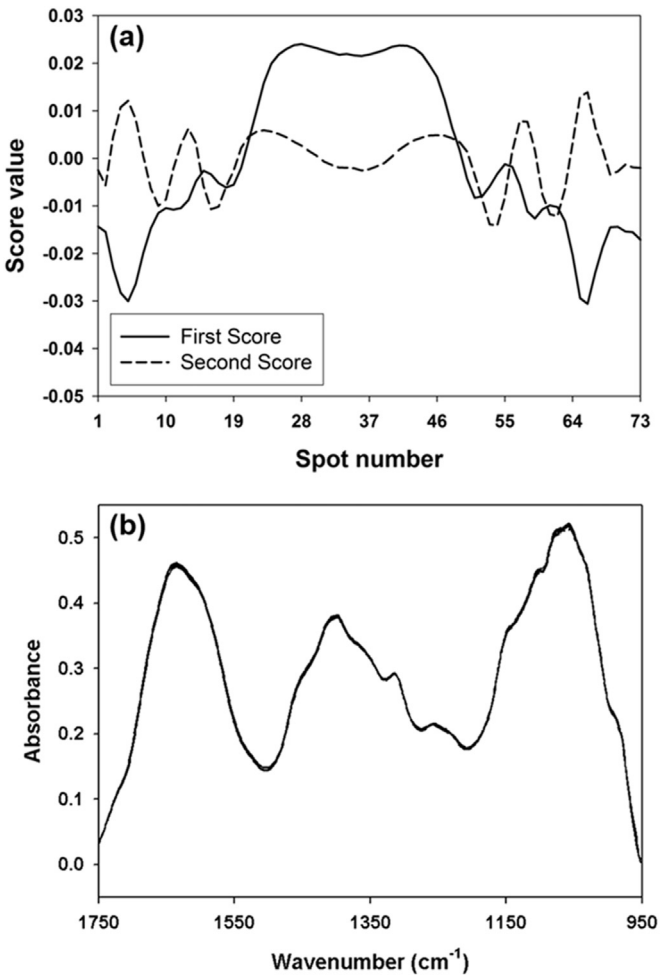

Fig. 4. Variation in the first (solid line) and second scores (dashed line) along the diameter of (a) the dried tobacco sample, and (b) IR spectra collected from the 8 dried tobacco samples shown in Fig. 1(j).

the spectral range, while the band shapes in the 1200–950 cm^{-1} range were relatively more dissimilar. Based on the literature, the bands in this range correspond to the combined features from sucrose, glucose and fructose [34]. Therefore, one of the differences between wild and transgenic great chloroplast tobaccos could be the amount and type of lower molecular weight carbohydrates. Probing the differences in metabolic pathways requires extensive full-component analytical data and an in-depth investigation of phytochemistry, which is beyond the scope of this study.

Using all of the sample spectra, PCA was performed, and the scores belonging to each group were compared. Fig. 5(b) presents a scatter plot constructed using the first and second scores. As shown, the groups of wild and transgenic tobaccos are discernible because the compositions of the extracted metabolites are dissimilar due to the differences in metabolic pathways. Based on these data we conclude that the drop-and-dry IR measurement using the FTS-coated CaF_2 substrate could be useful for metabolic fingerprinting where a reliable high throughput analysis is required.

4. Conclusion

We have demonstrated that various spectroscopic substrates with hydrophobic surfaces allow for the formation of reproducible droplets and consistent sample shapes after drying for reliable

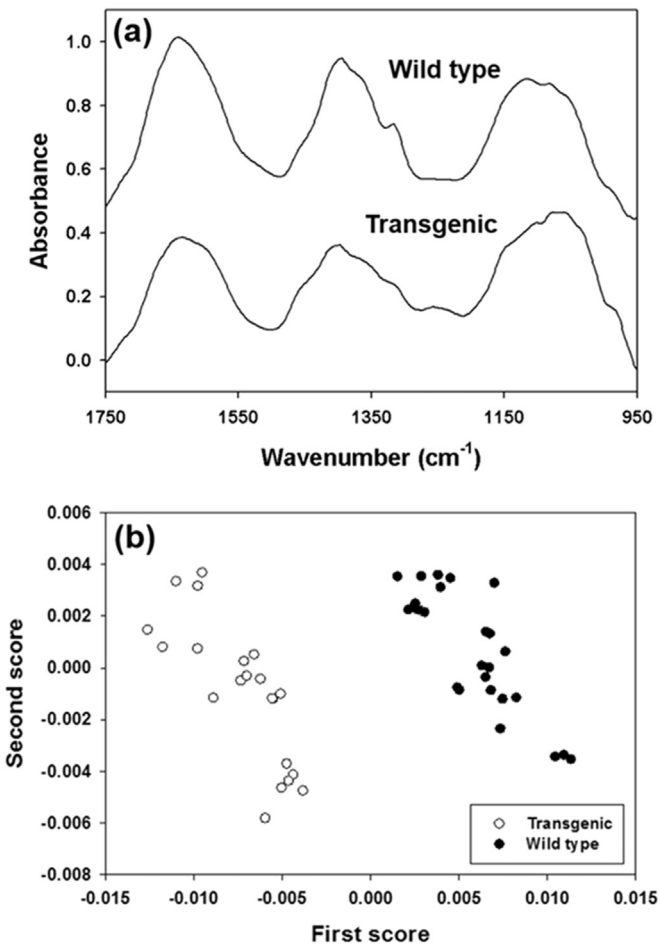

Fig. 5. (a) Average IR spectra of wild and transgenic chloroplast tobacco samples and (b) the scatter plot constructed with the first and second scores. Each spot is fully covered for the spectral acquisition.

drop-and-dry NIR and IR measurements. Depending on the concentrations of components in a sample, the use of a substrate with appropriate hydrophobicity was required. Specifically, the amine-coated substrate was proper for milk analysis, and the FTS-coated CaF_2 substrate was more effective to measure tobacco extracts. Therefore, when a new type of sample is subject to analysis, the search for optimal surface hydrophobicity that provides a reproducible spectroscopic measurement must be conducted first. Although the analyses for two different types of samples have been demonstrated here, the proposed scheme has potential for further spectroscopic analyses based on dry-n-drop measurements.

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